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GitHub Link	https://github.com/120608akhila/MLA0405-DEEP-LEARNING/upload/main/Assessment/CO2

SIMATS ENGINEERING

CO2 - ASSESSMENT TOOL 2

Scenario-Based Questions

COURSE NAME: Deep Learning
Faculty : Dr. Arul Raj A.M

COURSE CODE:MLA0405
Slot : D

COURSE OUTCOME COVERED

CO 2:Students will be able to apply supervised learning algorithms such as linear regression, classification models, decision trees, neural networks, support vector machines, and ensemble methods to solve real-world prediction and classification problems. They will gain the ability to select appropriate models, train them using datasets, and evaluate their performance. Students will also understand the strengths and limitations of different supervised learning approaches.(BTL-4)

CO Weightage : 15%

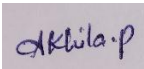
Assessment Tool Weightage: 40%

Duration: 90 minutes

Marks: 25

Code of Conduct:

I **Akhila PochimiReddy(192425225)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Name of the student: Akhila PochimiReddy

Scenario-Based Questions

CO-AT2

QUESTIONS: 05

Total Marks:25

1. Unsupervised Training of Neural Networks, Auto Encoders

A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.

Answer

Recommended Technique: Autoencoder (Unsupervised Neural Network)

Justification

Since the ECG dataset is **unlabeled**, an **Autoencoder** is the most suitable deep learning technique. It learns the normal characteristics of ECG signals by compressing and reconstructing the input data. When abnormal ECG signals are presented, the reconstruction error becomes significantly higher, allowing anomalies to be detected without labeled examples.

Workflow

1. Collect thousands of unlabeled ECG recordings.
2. Preprocess the signals by removing noise and normalizing the data.
3. Train an Autoencoder using the ECG recordings.
4. The encoder compresses the ECG into a low-dimensional latent representation.
5. The decoder reconstructs the original ECG signal.
6. Compute the reconstruction error.
7. ECG signals with reconstruction error above a threshold are classified as abnormal.
8. Cardiologists review the detected abnormal cases for diagnosis.

Advantages

- Does not require labeled ECG data.
- Learns meaningful latent representations automatically.
- Detects unknown and rare abnormalities.
- Reduces manual inspection effort.
- Improves early disease detection.

2. Auto Encoders, Representation Learning

A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation. Explain how an autoencoder can be used to solve this problem.

Answer

An **Autoencoder** is an unsupervised neural network used for **dimensionality reduction** and **representation learning**.

Working Process

1. Collect customer purchase information.
2. Normalize and preprocess the dataset.
3. Train the Autoencoder using customer purchase records.
4. The encoder compresses hundreds of purchase features into a smaller latent feature vector.
5. The decoder reconstructs the original purchase data.
6. The compressed feature vectors preserve important purchasing patterns while removing redundant information.
7. Apply clustering algorithms such as K-Means on the latent features.
8. Customers are grouped into meaningful segments such as frequent buyers, premium customers, seasonal buyers, and occasional shoppers.

Benefits

- Eliminates redundant features.
- Preserves important customer behavior patterns.
- Improves clustering accuracy.
- Reduces computational complexity.
- Enables better marketing strategies and personalized recommendations.

3. Width and Depth of Neural Networks, Activation Functions

A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

Answer

Recommended Changes

Increase Network Depth

- Replace the shallow neural network with a **deep neural network** containing multiple hidden layers.
- Deeper networks learn hierarchical facial features:
 - Layer 1: Edges
 - Layer 2: Eyes, nose, mouth
 - Layer 3: Complete facial structures
 - Final layers: Person identity

Improve Activation Functions

Use modern activation functions:

- **ReLU**
 - Faster convergence
 - Avoids vanishing gradients
 - Efficient computation
- **Leaky ReLU**
 - Prevents dead neurons.
 - Maintains small gradients for negative values.

- **Softmax**
 - Used in the output layer for multi-class face classification.

Advantages

- Better feature extraction.
- Improved learning capacity.
- Faster training.
- Higher recognition accuracy.
- Better generalization for unseen faces.

4. Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks

A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

Answer

A **Restricted Boltzmann Machine (RBM)** is an unsupervised probabilistic neural network used for collaborative filtering and recommendation systems.

Working Process

1. Create a user–movie interaction matrix.
2. Visible layer represents movies watched by users.
3. Hidden layer learns latent user preferences such as:
 - Action lovers
 - Comedy lovers
 - Romance lovers
4. Train the RBM using Contrastive Divergence.
5. The RBM learns hidden relationships between users and movies.
6. Predict ratings for unseen movies.
7. Recommend movies with the highest predicted preference scores.

Advantages

- Works without labeled data.
- Learns hidden user preferences.
- Handles sparse datasets effectively.
- Provides personalized movie recommendations.
- Improves recommendation accuracy.

5. Deep Learning Applications, Representation Learning

A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

Answer

Proposed Solution

Use a Deep Autoencoder with Representation Learning for defect detection.

Workflow

1. Collect sensor readings continuously from manufacturing machines.
2. Preprocess data by removing missing values and normalizing features.
3. Train the Autoencoder using normal product sensor data.
4. The encoder extracts compact latent representations capturing normal machine behavior.
5. The decoder reconstructs the sensor readings.
6. Calculate reconstruction error for every product.
7. Products with high reconstruction error are identified as defective.
8. Send alerts to quality control for inspection.

Advantages

- Detects defects automatically.
- Works with unlabeled sensor data.
- Identifies anomalies in real time.
- Reduces production waste.
- Improves product quality and manufacturing efficiency.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and	Mostly organized with minor structural	Some organization but lacks clear	No logical organization or structure

		grouping	issues	hierarchy	
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand